Joshua Alexander Conge

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EDUCATION

University of California, San Diego, La Jolla, CA

B.S Mechanical Engineering / M.S. Mechanical Engineering (Applied Mechanics) *September 2022 - June 2027*

- B.S. June 2026 | M.S. Expected June 2027
- Minor in Mathematics | GPA: 3.83/4.00
- Provost Honors, 3-Time Recipient

CERTIFICATIONS

California Board for Professional Engineers, Land Surveyors, and Geologists

Engineer-in-Training (EIT) - Mechanical Engineering *September 2026*

ENGINEERING EXPERIENCE

General Atomics - Electromagnetics, San Diego, CA

Manufacturing Project Engineer Intern *June 2026 - Present*

- Designed and developed a manufacturing fixture tool to support the inspection process, applying Design for Manufacturing (DFM) principles to improve manufacturability, usability, and repeatability.
- Applied GD&T principles and interpreted engineering drawings to establish critical dimensional and geometric requirements for accurate part positioning and inspection.
- Collaborated with manufacturing and engineering teams to identify inspection challenges and develop a fixture-based process improvement to increase inspection efficiency and consistency.

Vertical Flight Society, La Jolla, CA

Airframe Structures & Manufacturing Lead Engineer *October 2025 - April 2026*

- Led structural testing through Ansys FEA and manufacturing design of electric hexacopter capable of vertical takeoff and landing for wildfire response missions.
- Developed and implemented airframe fabrication and assembly processes, including composite layups and CNC machining.

UC San Diego MAE Senior Design Project, La Jolla, CA

Mechanical Engineering Student *September 2025 - June 2026*

- Developed a non-invasive medical monitoring device integrating spirometry, blood pressure, and ECG/heart-rate measurements for physiological assessment.
- Designed and implemented an ECG acquisition and processing system using an AD8232 sensor, Arduino, and custom signal-processing algorithms to extract heart rate and heart-rate variability (HRV).
- Implemented Pan-Tompkins-based ECG peak detection and RR-interval analysis in C++ to calculate heart rate and RMSSD, with automated handling of signal artifacts and lead-off conditions.

Triton Baja SAE, La Jolla, CA

Driver Interface & Safety (DIS) Engineer *September 2024 - October 2025*

- Engineered a mobile and sustainable pedal box design using Solidworks 3D Modeling & Drawing to ANSI Standard, reducing weight by 12%, optimizing it for off-road Baja SAE racing competitions.
- Fabricated pedal box, brake, and acceleration pedal using TIG welding and laser cutting techniques.
- Conducted structural analysis of the pedal box and chassis under various load conditions, utilizing Solidworks Finite Element Analysis (FEA) and MATLAB to evaluate deflection, yield strength, and material deformation.

UC San Diego Upper Division Housing, La Jolla, CA

Resident Assistant *August 2025 - June 2026*

- Built and maintained a safe, inclusive community for 60+ residents by planning and leading programs, mediating conflicts, and promoting academic and personal success.
- Served as a peer mentor, providing advice on academic, social, and wellness issues and connecting residents to appropriate campus resources.

KEY SKILLS & INTERESTS

Programming Languages: MATLAB, C++, Python, ROS2
CAD & Engineering: Solidworks, Onshape, Ansys, GD&T, DFM
Manufacturing & Fabrication: TIG Welding, Laser Cutting, Metal Bending, 3D Printing, CNC
Engineering Tools: VS Code, Microsoft Office Suite
Interests: Powerlifting, Swimming, Gaming, PC Building, Trading Card Collection, Reading